

Pediatrics and child health

OSCE 3

Down syndrome

- Pathogenesis

1)Non disjunction

2. Mosaic Down syndrome only in 1% of cases

3.Translocation Down syndrome....if there is advanced maternal age suspect this one

Clinical Features of Down Syndrome in the Neonatal Period

Central Nervous System

- Hypotonia*
 - Developmental delay
 - Poor Moro reflex*
 - Craniofacial
 - Brachycephaly with flat occiput
 - Flat face*
 - Upward slanted palpebral fissures*
 - Epicanthal folds
 - Speckled irises (Brushfield spots)
 - Three fontanel
 - Delayed fontanel closure
 - Frontal sinus and midfacial hypoplasia
 - Mild microcephaly
 - Short, hard palate
 - Small nose, flat nasal bridge
 - Protruding tongue, open mouth
 - Small dysplastic ears*
- Cardiovascular
 - Endocardial Cushings defects
 - Ventricular septal defect
 - Atrial septal defect
 - Patent ductus arteriosus
 - Aberrant subclavian artery
 - Pulmonary hypertension
- Musculoskeletal
 - Joint hyperflexibility*
 - Short neck, redundant skin*
 - Short metacarpals and phalanges
 - Short 5th digit with clinodactyly*

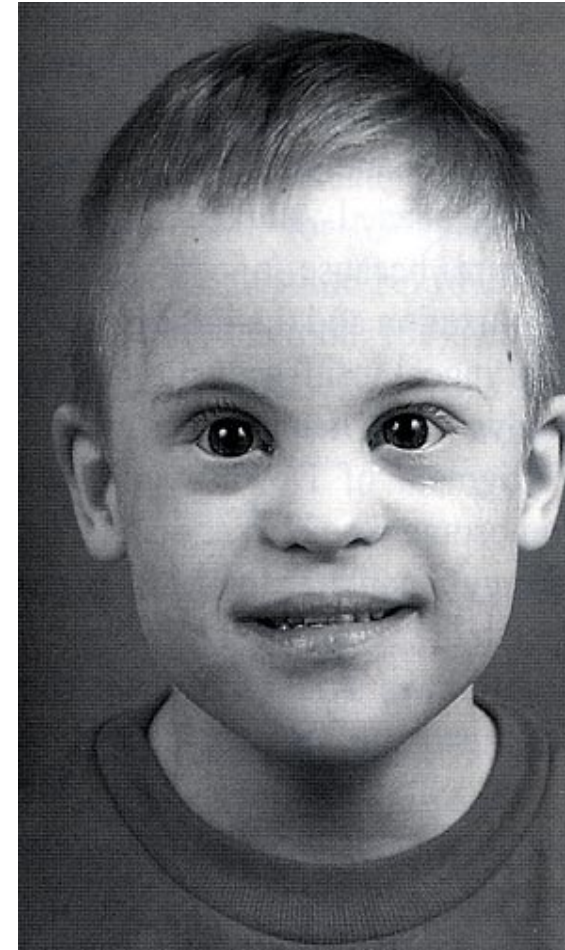
- Single transverse palmar creases*
- Wide gap between 1st and 2nd toes
- Pelvic dysplasia*
- Short sternum
- Two sternal manubrium ossification centers
- Gastrointestinal
- Duodenal atresia
- Annular pancreas
- Tracheoesophageal fistula
- Hirschsprung disease
- Imperforate anus
- Neonatal cholestasis
- Cutaneous
- Cutis marmorata
- *

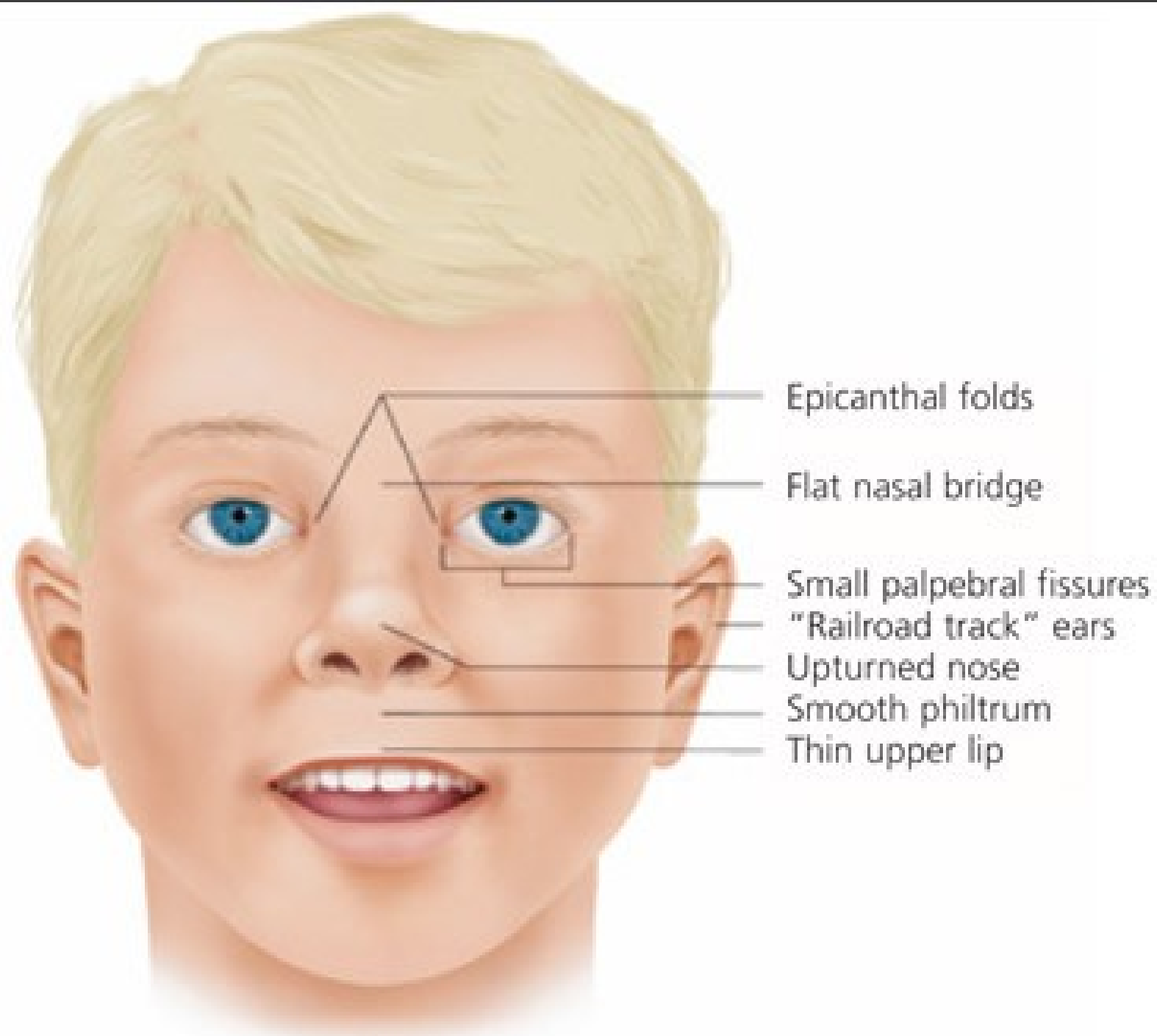
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- **Hall's criteria to aid in diagnosis.**
- **Presence of the above dysmorphic feature**
- **If #dysmorphic features >8=down syndrome**
- **If #dysmorphic features >5=mosaic down syndrome**

Down syndrome

- Facial Characteristics
 - Macroglossia
 - Micrognathia
 - Midface hypoplasia
 - Flat occiput
 - Flat nasal bridge
 - Epicanthal folds
 - Up-slanting palpebral fissures
 - Progressive enlargement of lips





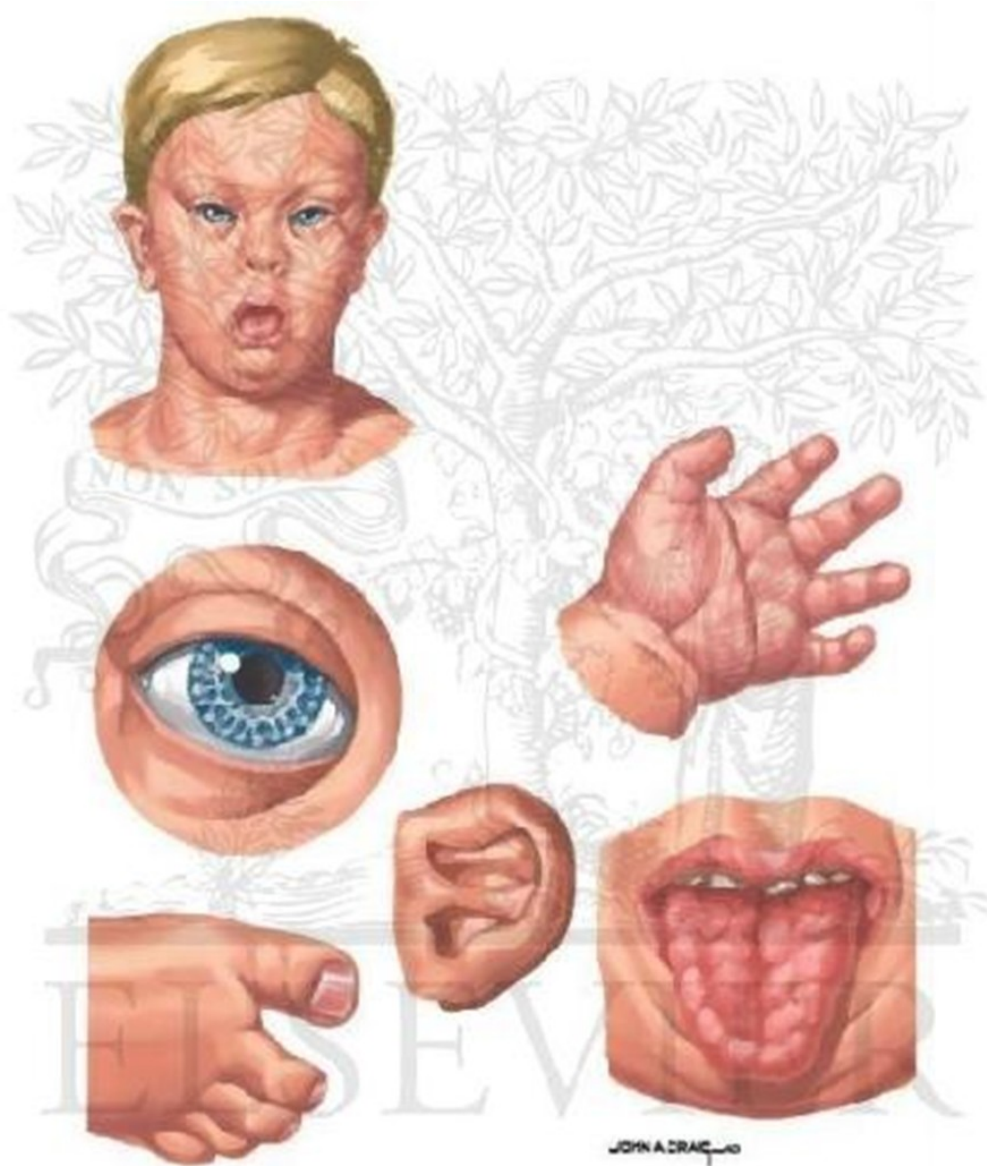
Brushfield spots



- small white or grayish/brown spots on the periphery of the iris

Extremities

- Short broad hands
- Incurved fifth finger with hypoplastic mid phalanx
- Transverse palmar crease
- Space between the first and second toes (sandal gap)
- Hyperflexibility of joints



Trisomy 21, Down syndrome



...Trisomy 21, Down syndrome



- Hypotonia

Trisomy 21, Down syndrome

- ▶ flat face
- ▶ upward and slanted palpebral fissures
- ▶ epicanthic folds
- ▶ speckled irises (Brushfield spots)
- ▶ flat nasal bridge
- ▶ small mouth with protruding tongue



....Trisomy 21, Down syndrome

- Single palmar crease, simian crease
- short, broad hands
- hypoplasia of middle phalanx of 5th finger





...Trisomy 21, Down syndrome



- Pronounced 'sandal' gap between the big and second toe

Marfan Syndrome

- Autosomal dominant
 - FBN 1 gene encoding extracellular matrix protein – Fibrillin 1
 - Traditionally thought problem with elastic fiber deposition (affects aorta, skin)
- Occurs 1/5000 – 1/10,000 live births
 - 30% sporadic d/t new mutations
- Primarily affects
 - Skeletal
 - Cardiovascular
 - Ocular

Skeletal system

- Chest deformity (pectus carinatum or excavatum)
- Long arms
- Long fingers and loose



- Scoliosis
- Joint laxity and hypermobility – can eventually lead to contractures



Patau's syndrome

Trisomy 13

Major Clinical Features

- mental retardation
- growth retardation
- microcephaly
- cleft lip/palate
- small jaw (micrognathia)
- deformed, low-set ears
- Fingers flexed with postaxial polydactyly
- congenital heart defects
- seizures
- low birth weight

Early lethality in most cases, with a median survival of 7 days; 91% die by 1 year



Patau's syndrome – Trisomy-13

Patau synd...



Patau synd...polydactyly



Edward's syndrome

Trisomy 18 (Edwards)

Major Clinical Features

- mental retardation
- Low birth weight
- Short sternum
- Microcephaly
- Narrow and dislocated hips/abnormal pelvis
- hypertonia
- congenital heart disease
- horseshoe kidneys
- hydronephrosis
- short sternum
- pyloric stenosis

Most die in utero, otherwise with 1st few days

Edwards syndrome - prominent occiput, microcephaly, micrognathia, low-set ears, short sternum, narrow pelvis, prominent calcaneus, and flexion abnormalities of the fingers, congenital heart disease, horseshoe kidney





- overlapping fingers;
index over 3rd, 5th over
4th; closed fist

- Rocker bottom feet



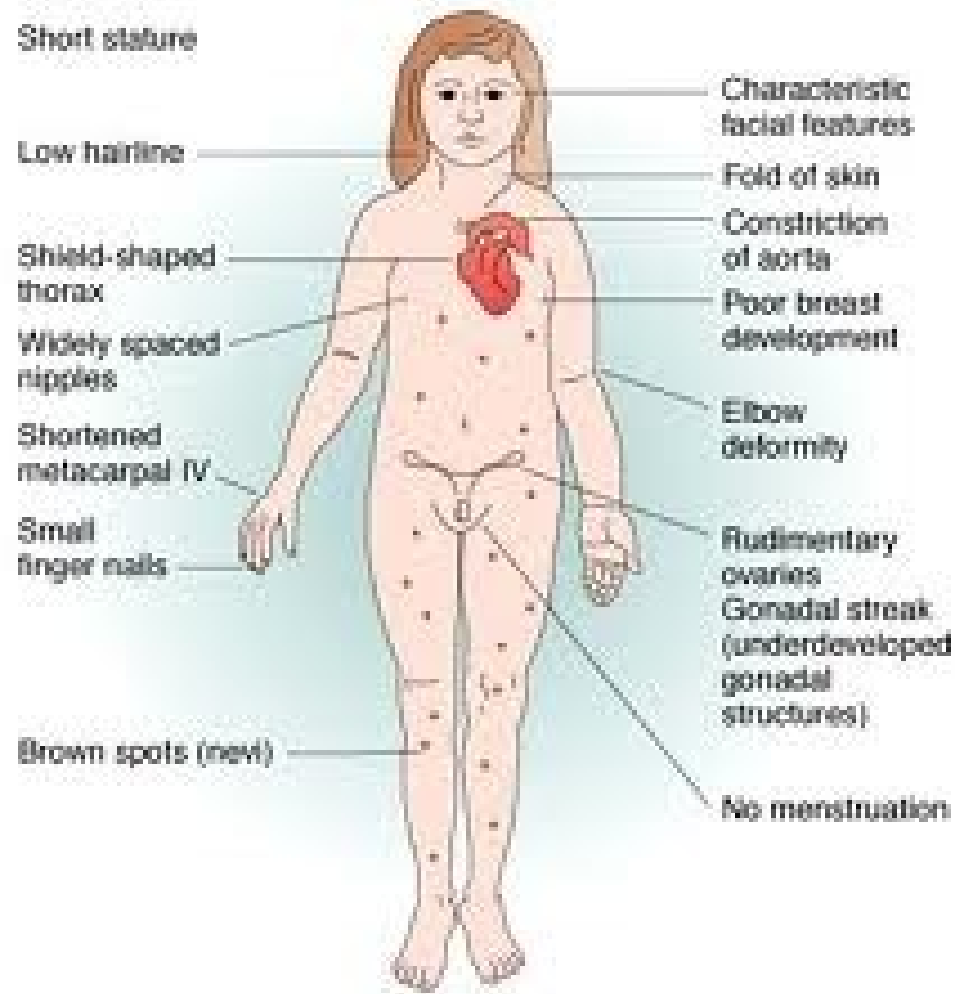
Turner's Syndrome

Major Clinical Features

- Occurs 1/2000-1/5000 females
- female phenotype
- Short stature (less than 5 feet)
- **primary amenorrhea**
- low estrogen levels
- maldevelopment of the ovaries
- Sterility
- Horseshoe kidney
- webbing of the skin of the neck
- wide-spaced nipples
- edema at birth
- cardiovascular problems
 - Coarctation of aorta
 - Bicuspid aortic valve
- Hypothyroidism
- Diabetes
- Lymphadema (swelling of hands and feet)



Turner syndrome – clinical features



TURNER SYNDROME

CLINICAL MANIFESTATIONS[3]

abnormal ears and
micrognathia



lymphedema of
hands and feet



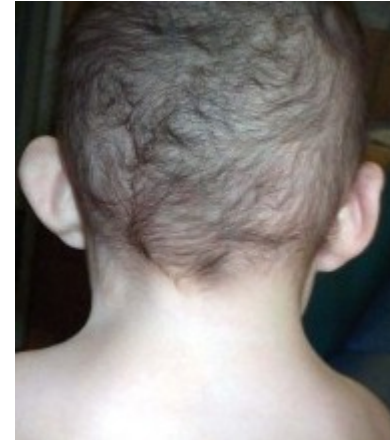
TURNER SYNDROME

CLINICAL MANIFESTATIONS[4]

shield chest with
widespread nipples

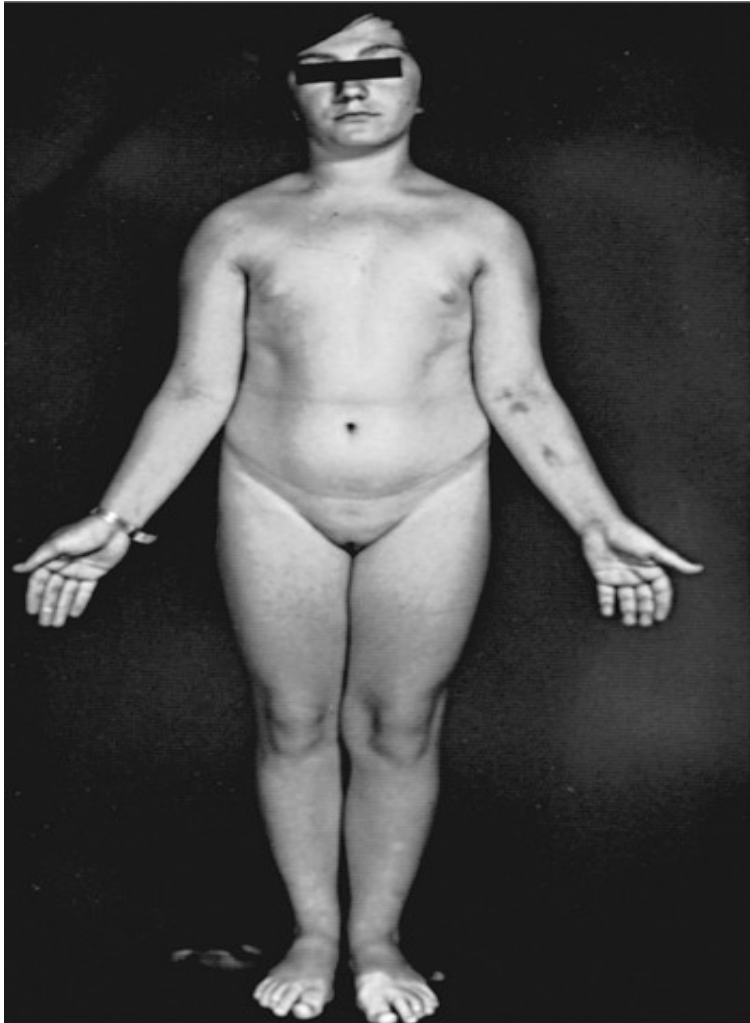


webbed neck with low
hairline



TURNER SYNDROME

CLINICAL MANIFESTATIONS[5]



- short stature
- cubitus valgus

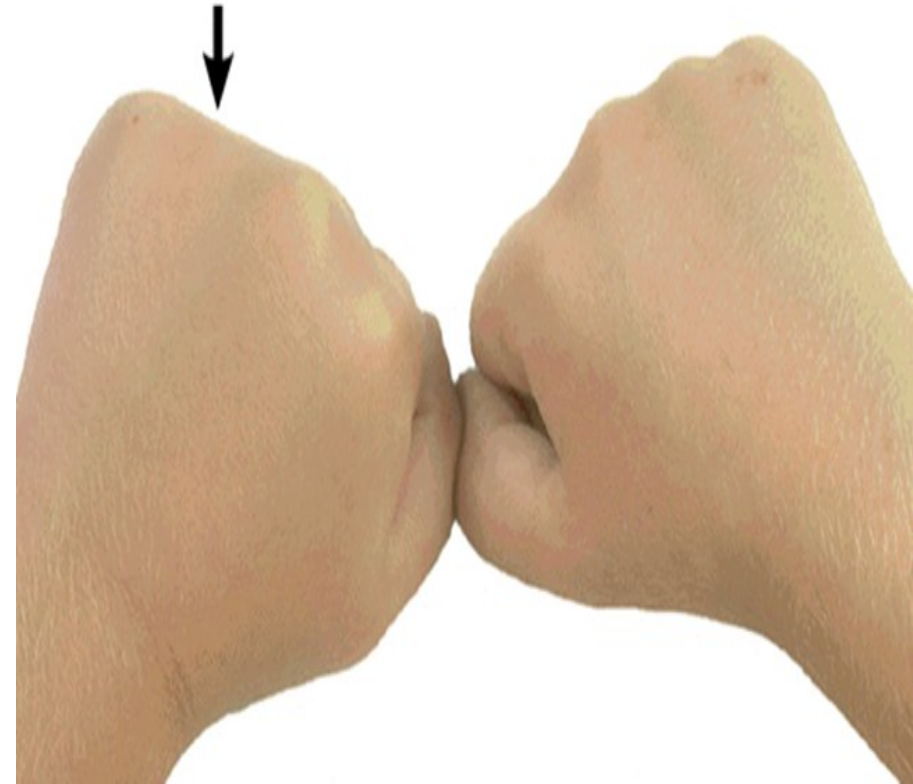
TURNER SYNDROME

CLINICAL MANIFESTATIONS[6]

Nail dysplasia, present
in 75%



Short 4th metacarpals



Turner Syndrome

CLINICAL MANIFESTATIONS[7]



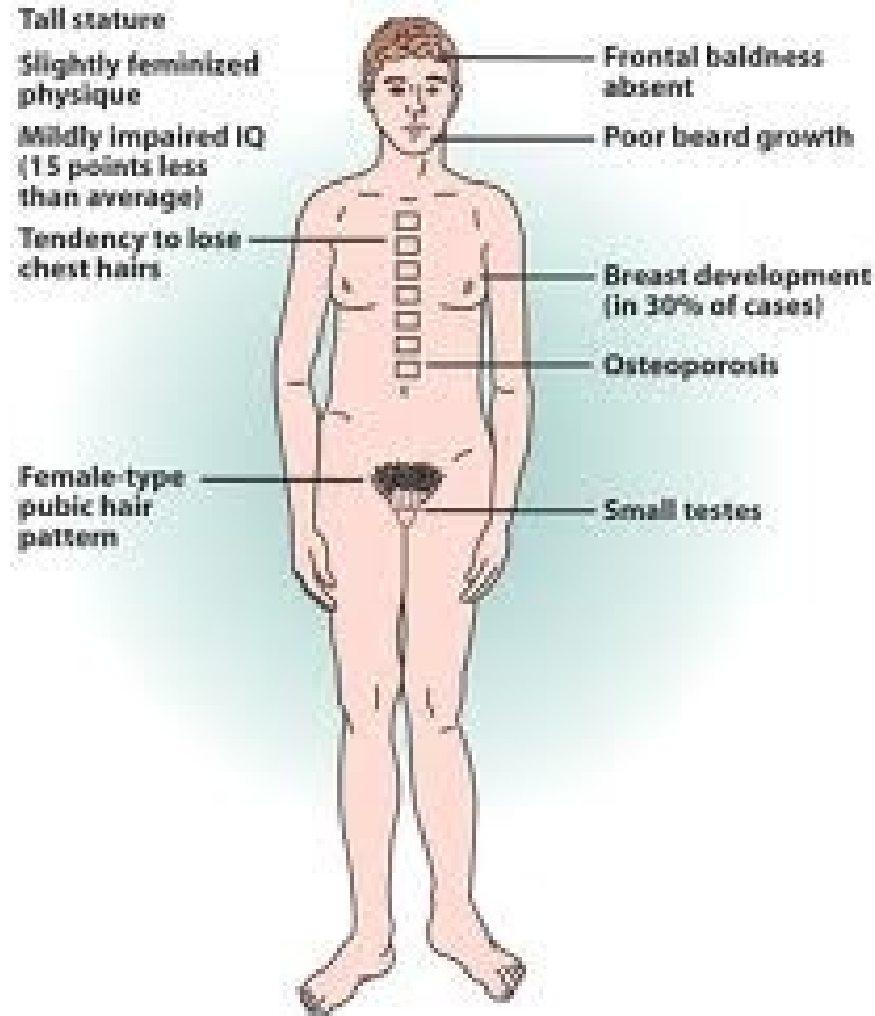
- Madelung deformity

Klinefelter's Syndrome

Major Clinical Features

- Most common sex chromosomal abnormalities in males (1/500-1/1000)
- Rare forms: 48 XXXY
- Mosaic 47 XXY, 46 XX
- Usually after puberty(if not prenatally)
 - do not produce as much testosterone as other boys, they have a less muscular body, less facial and body hair, and broader hips
- Varying mental retardation in some
- Social impairment
- Most have generalized learning disabilities, usually language based
- Tend to be **tall** and slim with long limbs, large hands and feet
- **Small testes**
- **aspermia** (little to no sperm production), Hypogonadism
 - fertility in some
- Gynecomastia during adolescence

Clinical features cont...



- Diagnosis and treatment:
- Karyotype of peripheral leukocytes
- Testosterone replacement

DiGeorge Syndrome...

- Variable hypoplasia of the thymus & parathyroid glands- partial DiGeorge syndrome- More frequent
- Total aplasia -- complete DiGeorge syndrome.
- 15% are born to diabetic mothers.
- 15% of infants have no identified risk factors.
- 30% of infants with complete DiGeorge syndrome have CHARGE association .

Clinical features

- Children with partial thymic hypoplasia may have little trouble with infections & grow normally.
- Patients with complete DiGeorge syndrome resemble patients with SCID in their susceptibility to infections.
- Low-grade or opportunistic pathogens:
 - Fungi, viruses & Pneumocystis jiroveci
 - GVHD from nonirradiated blood transfusions.

RX - Thymic tissue transplants
- Bone marrow or peripheral blood transplantation.

DiGeorge syndrome

- Thymic and parathyroid glands hypoplasia
results from dysmorphogenesis of the
3rd and 4th pharyngeal pouches

- The **classic triad** of features of DGS on presentation is
 - conotruncal cardiac anomalies
 - hypoplastic thymus
 - hypocalcemia (resulting from parathyroid hypoplasia)

- The dysmorphic features include
 - Hypertelorism, antimongloid slant
 - Low set notched ears
 - short philtrum of the upper lip
 - Mandibular hypoplasia
- occurs in both males and females

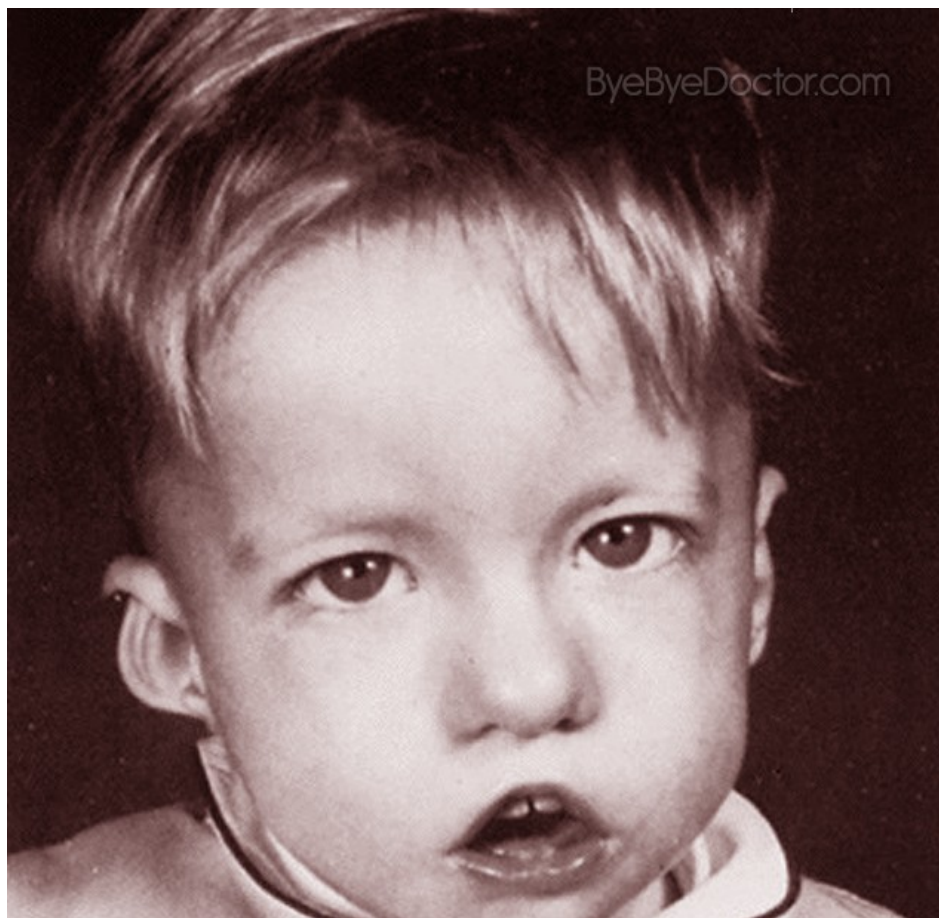
- The immune defects in DGS include
 - impaired cellular immunity
 - CD4+ T-cell count of < 1000 cells/mm³

Boy 14months
Bronchopneumonia ,CHD
Immunodeficiency
Hypocalcemia
Nearside facial paralysis



- It is critical to confirm the diagnosis of complete DGS
- Because it is fatal without treatment.
- The diagnosis is 1st suggested by hypocalcemic seizures during the neonatal period
- Treatment
 - Bone marrow transplantation.

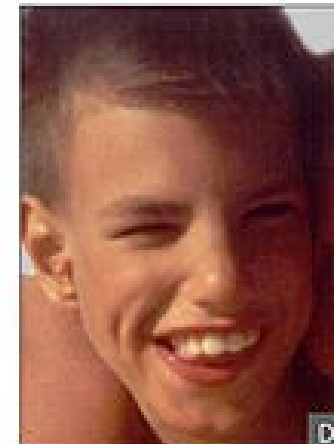




Cri du chat

Major Clinical Features

- Distinctive cat-like cry 1st few weeks of life
- Profound developmental retardation
- Severe mental retardation
- Microcephaly
- Hypotonia
- Hypertelorism
- Epicanthal folds
- High arched palate
- Wide and flat nasal bridge

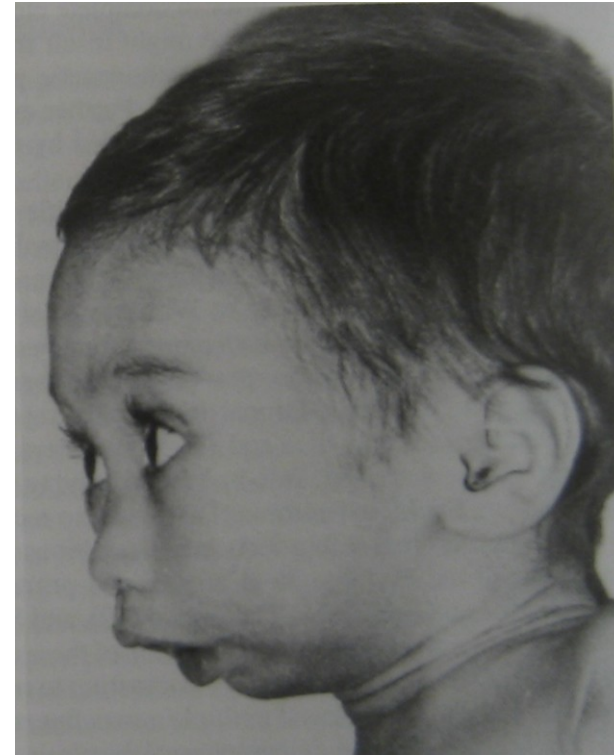
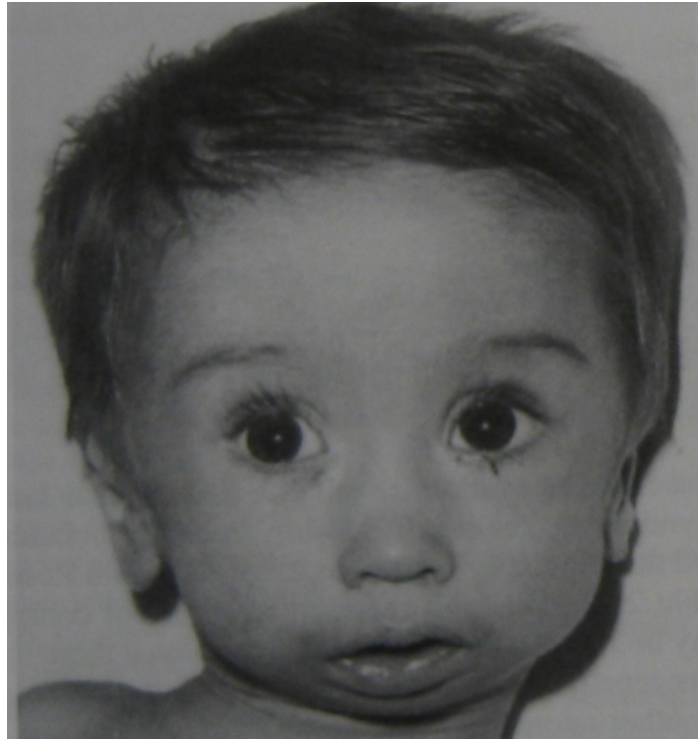


Pierre Robin Sequence

- Triad of cleft palate, micrognathia, and glossoptosis (airway obstruction caused by backwards displacement of the tongue base)
- 1923, credit to French stomatologist
- Incidence 1 in 8500



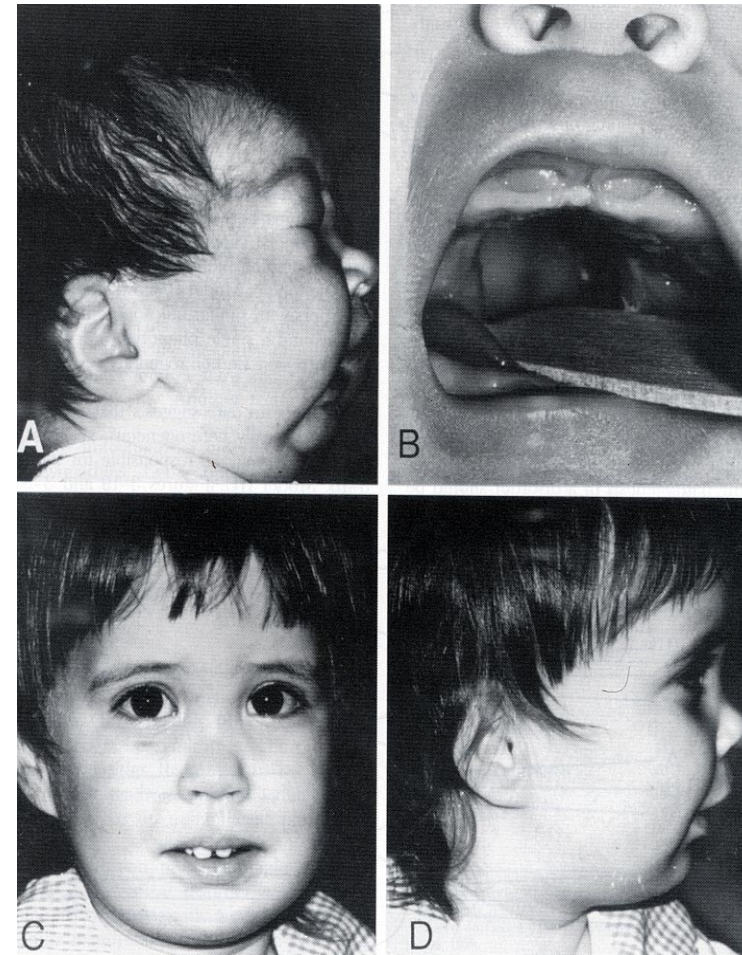
PRS



Pictures From: Gorlin et al: *Syndromes of the Head and Neck*. New York, Oxford University Press, 1990.

PRS (cont.)

- Mandibular deficiency
- Etiology can be multiple (positional, genetic, neurologic, connective tissue d/o)
- Nonsyndromic (80%) or syndromic (20%)



Treacher-Collins Syndrome

- Autosomal dominant
- 1 in 25,000 –50,000 births
- Bilateral abnormalities of 1st and 2nd branchial arches
- Hypoplasia of maxilla, zygoma, and mandible
- Downward slanting eyes with colobomas of lower eyelid and absence of eyelashes
- Normal intelligence

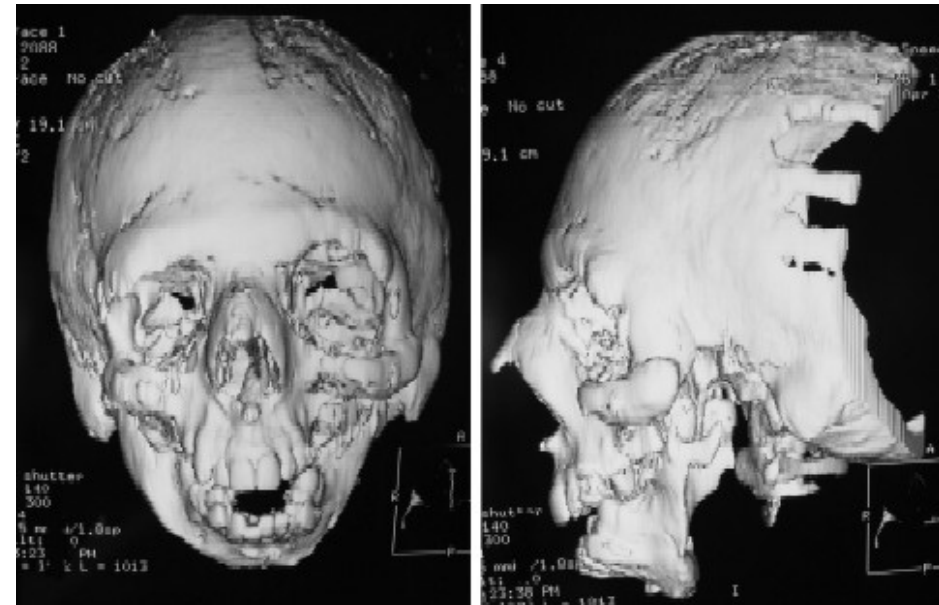


TCS



Picture From: Cummings, CW: *Otolaryngology: Head and Neck Surgery*. St Louis, Mosby, 1998

TCS



Picture From: Acosta, HL et al: Vertical Mesenchymal Distraction and Bilateral Free Fibula Transfer for Severe Treacher Collins Syndrome. *Plastic & Reconstructive Surgery*, 113(4), 2004.

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